

No. of Functional Features Included in the Solution

|               |                          |
|---------------|--------------------------|
| Date          | 15 March 2024            |
| Team ID       | 6aba3aa9ca93a27509644a39 |
| Project Name  | ComicCraft               |
| Maximum Marks | 5 Marks                  |

Functional Features Overview:

This document lists the functional features included in the ComicCraft project solution. The features are derived from the project description, scenarios, architecture, workflow, backend routes, AI integration, frontend templates, and deployment/testing details in the supplied project document.

| S.No | Feature Name                         | Feature Description                                                                         | Module / Component | Status (Done / In Progress / Pending) | Marks Contribution |
|------|--------------------------------------|---------------------------------------------------------------------------------------------|--------------------|---------------------------------------|--------------------|
| 1    | Story Prompt Input                   | Collects the user's main comic idea through the web form.                                   | Frontend / UI      | Done                                  | 0.50               |
| 2    | Character, Setting, Tone & Art Style | Captures character name, setting, story tone, and preferred art style for personalization.  | Frontend / UI      | Done                                  | 0.50               |
| 3    | AI Comic Outline Generation          | Gemini Flash generates a structured 5-panel comic outline with scene and image prompts.     | AI / Backend       | Done                                  | 0.75               |
| 4    | Narration & Dialogue Generation      | Gemini Pro expands the outline into detailed narration and character dialogue.              | AI / Backend       | Done                                  | 0.75               |
| 5    | AI Comic Illustrations               | Stable Diffusion generates comic-style illustrations from the image prompts.                | AI / Image         | Done                                  | 0.75               |
| 6    | Comic Preview & Layout               | Displays generated panels sequentially and organizes images, titles, descriptions and text. | Frontend / Backend | Done                                  | 0.50               |

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|------|---------------------------------|--------------------------------------------------------------------------------------------|--------------------|---------------------------------------|--------------------|
| 7    | PDF Export                      | FPDF compiles panel images and narrative text into a downloadable multi-page PDF.          | Backend / Export   | Done                                  | 0.75               |
| 8    | FastAPI API & Route Integration | Handles form/JSON generation, image testing, export flow and coordination of AI functions. | Backend / API      | Done                                  | 0.50               |

**Feature Summary:**

| Metric                             | Count / Value                                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Total Features Planned             | 8                                                                                                                     |
| Total Features Implemented         | 8                                                                                                                     |
| Core / Must-Have Features          | Story input, AI outline, narration/dialogue, illustrations, preview/layout and PDF export                             |
| Additional / Nice-to-Have Features | Customization of character, setting, tone and art style; API access and image-testing route                           |
| Features Tested & Verified         | Local deployment and complete comic-generation workflow are described as tested and verified in the project document. |

**Prepared by: Tarunikha**

**Feature Category Breakdown:**

| S. No | Category                  | Features in Category | Example Features                                                                                                                                     |
|-------|---------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | User Interface (UI)       | 2                    | Story prompt input; character/setting/tone/art-style selection; comic preview                                                                        |
| 2     | Backend / Logic           | 3                    | FastAPI routing; workflow coordination; layout building; PDF export                                                                                  |
| 3     | Database / Storage        | 0                    | No database or user-account storage is described in the supplied project document                                                                    |
| 4     | API / Integration         | 4                    | Gemini Flash; Gemini Pro; Stable Diffusion/Diffusers; FastAPI JSON API                                                                               |
| 5     | Security / Authentication | 0                    | No user authentication feature is described; the conclusion states the current version focuses on single comic generation without account management |

**Project Details Used to Fill the Form:**

- The application is a web-based AI comic creator using FastAPI, Google Gemini models and Stable Diffusion.
- Inputs include story prompt, main character name, setting, tone and art style.
- The workflow generates a structured comic outline, narration/dialogue, illustrations, a panel layout and a downloadable PDF.
- FastAPI routes include /generate, /generate-comic/json, /export-success and /test-image.
- The frontend uses HTML, CSS and Jinja2 templates including index.html, comic\_preview.html and export\_success.html.
- The project document describes local deployment and testing using Uvicorn and the FastAPI application.

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